Sibling points Stage 4b: aut=3 re-banded

Sibling points, Stage 4b: the aut=3 panel re-banded with its canonicalisations priced (results, local L0)

What ran. The frozen panel of `PROTOCOL.md` (commit `709f978290`, frozen before data). It re-runs Stage 4's design on fresh seeds and curves, with the engine on `main` that lists the `aut=3` canonicalisations as `aut_canonicalisations_uncharged`. That is the fix for the Stage 4 validator's erratum E1. The binary was built from `4e2ea6c1be`.

The panel used j = 0, b = 7 prime-order curves (the secp256k1 shape) at p = 34, 37 and 40 bits, with five curves per size, 6 targets per curve and 6 rounds plus 1 warmup. The arms were rho-neg, rho-cubic, rho-neg-aa, tw2cj-lazy, tw2cj-lazy-aut3, sib8-p512 and sib8-p512-aut3. It ran locally at L0 under bench record `benchmarks/records/20261112T0311Z-sib-stage4b-aut3.json`.

The analysis is native only: `compare.sh` calls `ecbench compare` (GAE at w = 0, 0.1 and 1, and `--unit bitop/v1`), `ecbench band` and `ecbench trend`. The saved comparison files under `sessions/*/comparisons/` are the results; `compare.txt` is the run log. All numbers are `[pcc:unverified]`, with no pcc claim filed.

Admission. All 3 specs pass: 3 × 1,470 records verified, `verify --replay 8` reproduced 8 of 8 per spec with 0 problems, and the A/A ratio is exactly 1 at every weight [pcc:unverified].

Results: paired GAE ratios, 180 pairs per cell (ecbench compare)

Each cell gives the ratio at w = 0 / 0.1 / 1. The bracket is the 95 percent interval at w = 0.

ratio	p 34	p 37	p 40	band order
sib8-p512-aut3 / rho-cubic	**0.992 [0.902, 1.092]** / 1.356 / 3.000	**0.875 [0.804, 0.952]** / 1.200 / 2.663	**0.909 [0.828, 0.997]** / 1.248 / 2.776	unresolved at every size
tw2cj-lazy-aut3 / rho-cubic	1.846 / 2.014 / 2.770	1.805 / 1.968 / 2.707	1.911 / 2.085 / 2.866	rho-cubic lower at every weight
sib8-p512 / rho-cubic	1.680 / 2.001 / 3.450	1.544 / 1.840 / 3.175	1.495 / 1.782 / 3.078	rho-cubic lower at every weight
tw2cj-lazy-aut3 / tw2cj-lazy	0.560 / 0.611 / 0.838	0.563 / 0.614 / 0.844	0.575 / 0.627 / 0.862	aut3 lower at every weight
sib8-p512-aut3 / sib8-p512	0.590 / 0.678 / 0.870	0.567 / 0.652 / 0.839	0.608 / 0.700 / 0.902	aut3 lower at every weight
rho-cubic / rho-neg	0.556	0.544	0.558	rho-cubic lower at every weight

Bit-ops (`ecbench compare --unit bitop/v1`): sib8-p512-aut3 / tw2cj-lazy is 0.368 / 0.334 / 0.334. - The bitop/v1 instrument question persists for rho-cubic: rho-cubic / rho-neg is 1.059 / 1.041 / 1.072 in bit-ops, against 0.55 in GAE. - So sib8-p512-aut3 / rho-cubic in bit-ops (0.621 / 0.547 / 0.568) is not used.

Slope in r (`ecbench trend`, fit `free_r`, w = 0 comparisons over the three sizes): sib8-p512-aut3 / rho-cubic is -0.015 [-0.051, +0.021].

Stage 4b

Against the pre-registered expectations

- **J1 holds.** At w = 0, sib8-p512-aut3 / rho-cubic is 0.992, 0.875 and 0.909, inside [0.80, 1.00]. At p 34 it sits at the top of the range, with an interval that crosses 1.
- **J2 holds.** At w = 0.1 the ratio is 1.356, 1.200 and 1.248, inside [1.10, 1.40]. At w = 1 it is 3.000, 2.663 and 2.776, inside [2.4, 3.1]. The band is unresolved at every size.
- **J3 holds.** aut3 / non-aut3 is 0.56 to 0.61 at w = 0, inside [0.50, 0.66], and 0.84 to 0.90 at w = 1, inside [0.70, 0.95].
- **J4 holds.** Every one of the 1,080 measured aut3 records lists `aut_canonicalisations_uncharged` in `cost.unpriced`, and its count equals the record's `table_insert_attempts`.

What this means

- **The validator's corrected band reproduces on fresh curves.** The aut=3 sibling arm is at or below rho-cubic at w = 0 (0.88 to 0.99). It is 1.20 to 1.36 times rho-cubic at w = 0.1 and 2.7 to 3.0 times at w = 1, flat in r. It is unresolved in the band, and rho-cubic stays well ahead once native operations carry any weight [pcc:unverified].
- **At w = 1 the sibling mechanism no longer pays** against the plain aut=3 walk. tw2cj-lazy-aut3 is 2.77 / 2.71 / 2.87 times rho-cubic, against 3.00 / 2.66 / 2.78 for sib8-p512-aut3: lower at p 34, about level at p 37 and p 40. This is the design note's arithmetic (`../stage5-design/README.md`): two inserts and a second point per pair.
- **Not claimed.** No below-rho result in the band, no exponent statement, no wall-time result, no leaderboard row, no claim about secp256k1.

Next

An independent validator re-checks this directory. If it passes, `VALIDATION.json` opens the gate on these numbers.

Files: - PROTOCOL.md, frozen.sha256, spec-j0b7-{34,37,40}.json - run_local.sh, compare.sh, compare.txt - sessions/ (records, saved comparisons), trend/ - verify-*.json, band-*.{json,txt} - plot.py (renders the saved comparisons; computes nothing), file:///home/user/crypto/research/curve25519_autolab_20261104/siblings/stage4b-aut3/figures/, REPORT.pdf